

Kernel-Agnostic Gradient-Matched Position Step

Let agent i have influence kernel $k(x_i, b) > 0$, and define

$$f_i(b) := k(x_i, b), \quad G_i(\mathbf{x}, b) := \frac{f_i(b)}{\sum_j f_j(b)}.$$

Its utility is

$$u_i(\mathbf{x}) := \sum_b B(b) G_i(\mathbf{x}, b).$$

Define the score of the kernel with respect to position:

$$s_i(b) := \nabla_{x_i} \log k(x_i, b).$$

Then

$$\nabla_{x_i} G_i(b) = G_i(b) (1 - G_i(b)) s_i(b),$$

and therefore

$$\nabla_{x_i} u_i = \sum_b B(b) G_i(b) (1 - G_i(b)) s_i(b).$$

Position-only style Monte-Carlo estimator

Under canonical routing $p_i(b) \propto B(b)G_i(b)$, samples routed to agent i are distributed according to that density. For routed samples $\{b_m\}_{m \in \mathcal{Q}_i}$, define

$$w_m := 1 - G_i(b_m), \quad d_i := \frac{\sum_{m \in \mathcal{Q}_i} w_m s_i(b_m)}{\sum_{m \in \mathcal{Q}_i} w_m + \varepsilon}.$$

Because routing contributes one factor G_i , and w_m contributes the factor $(1 - G_i)$, we have

$$\mathbb{E}[d_i] \propto \sum_b B(b) G_i(b) (1 - G_i(b)) s_i(b) = \nabla_{x_i} u_i.$$

So d_i is gradient-matched in expectation for any differentiable kernel.

Generic update rule

Use

$$x_i^+ = x_i + \eta P_i d_i,$$

where $\eta > 0$ is step size and $P_i \succ 0$ is an optional preconditioner (take $P_i = I$ by default).

Gaussian special case

For multivariate Gaussian kernels,

$$s_i(b) = \Sigma^{-1}(b - x_i),$$

which recovers the earlier position-only centroid form up to metric and normalization factors.